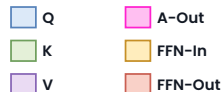


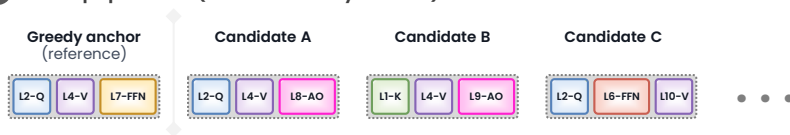
(b) Inside the genetic search — example $k = 3$

1 Representation

An individual is a set of LoRA modules selected to receive the next k rank components.
Each tag encodes a layer–module pair; search spans all eligible target modules.



2 Initial population (around Greedy anchor)



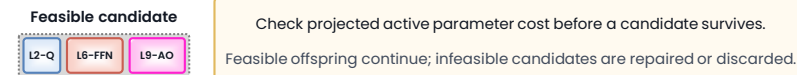
3 Selection & Set crossover (elitism picks parents)



4 Mutation (add / remove / replace)



5 Budget repair (ensure feasibility)



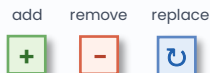
Genetic operators



Set crossover

Exchange subsets between parents

Mutation



Elitism

Keep best individuals



Diversity

Hamming distance between sets



Repeat for several generations

Shortlist feasible, diverse candidates for each k

Send shortlist to training-only calibration and final gating